Single Dimensional Array

Address Calculation in a Single Dimension Array:

Example 2: An Array A [-15------ 64] is stored in computer memory whose base address is 459, word size is 2 byte.

a. **How many numbers of elements are there?**

Solution:

No of Elements = Size of Array

= $\mathbf{U_b - L_b + 1}$

= 64 - (-15) +1

= 64+15+1

= 80.

b. **Total memory size?**

Solution:

Total memory size = Size of Array * word size

= 80*2

= 160.

c. **Find memory location of A [10].**

Solution:

Address of A [I]= B + W*(I-L_b)

Here,

B = 459

W= 2 bytes

I = 10

L_b = -15

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Address of A[10] = 459+2*(10 – (-15))

= 459 + 2*(10+15)

= 459 + 2*(25)

= 459 +50

= 509.

d. Which element is located at memory address 589?

Solution:

Address of A [I]= B + W*(I-L_b)

Here, we find I.
589 = 459+2*(I – (-15))
130 = (I+15)*2
65 = I+15
I= 50.

Practice Question

1. Consider the linear array A[-10;10], B[1925:1990], C[25].

(a) Find the number of elements in each array

(b) Suppose base (A)=400 and word size (A)=2 words, find the addresses of A[-3], A[0], A[3]

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